



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# **AI-Based Smart City Framework for Crime Prediction and Intelligent Emergency Response**

**CSA0927 – PROGRAMMING IN JAVA**

**Guided by:  
S. Puspa Latha**

**PRESENTED BY**

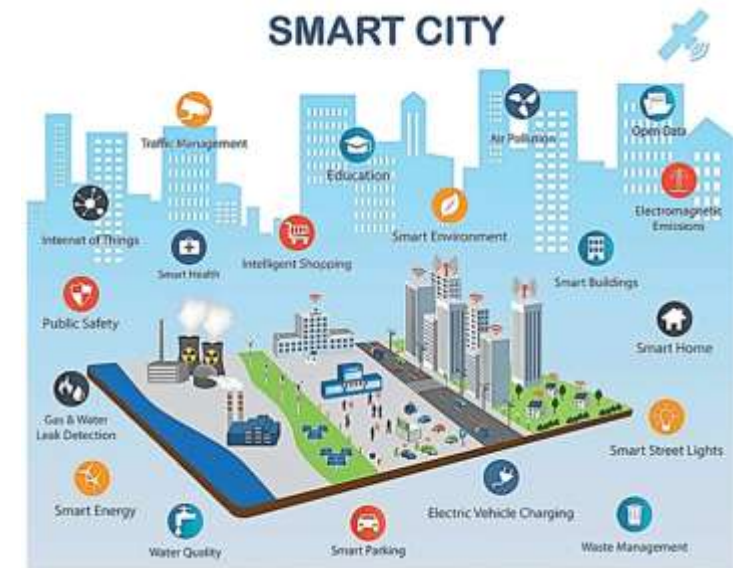
**C.DEVASRI(192521422)**

**N.RAMYA(192511403)**

**K.SRITHIKA(19254448)**

# Introduction

- SafeCity AI is an AI-powered public safety system that predicts crime-prone areas using historical crime data.
- The system analyzes crime patterns to help police take preventive actions before crimes occur.
- It provides a quick emergency response feature that allows citizens to report incidents instantly.
- The main aim is to reduce crime, improve response time, and create a safer environment for everyone.



# Problem statement

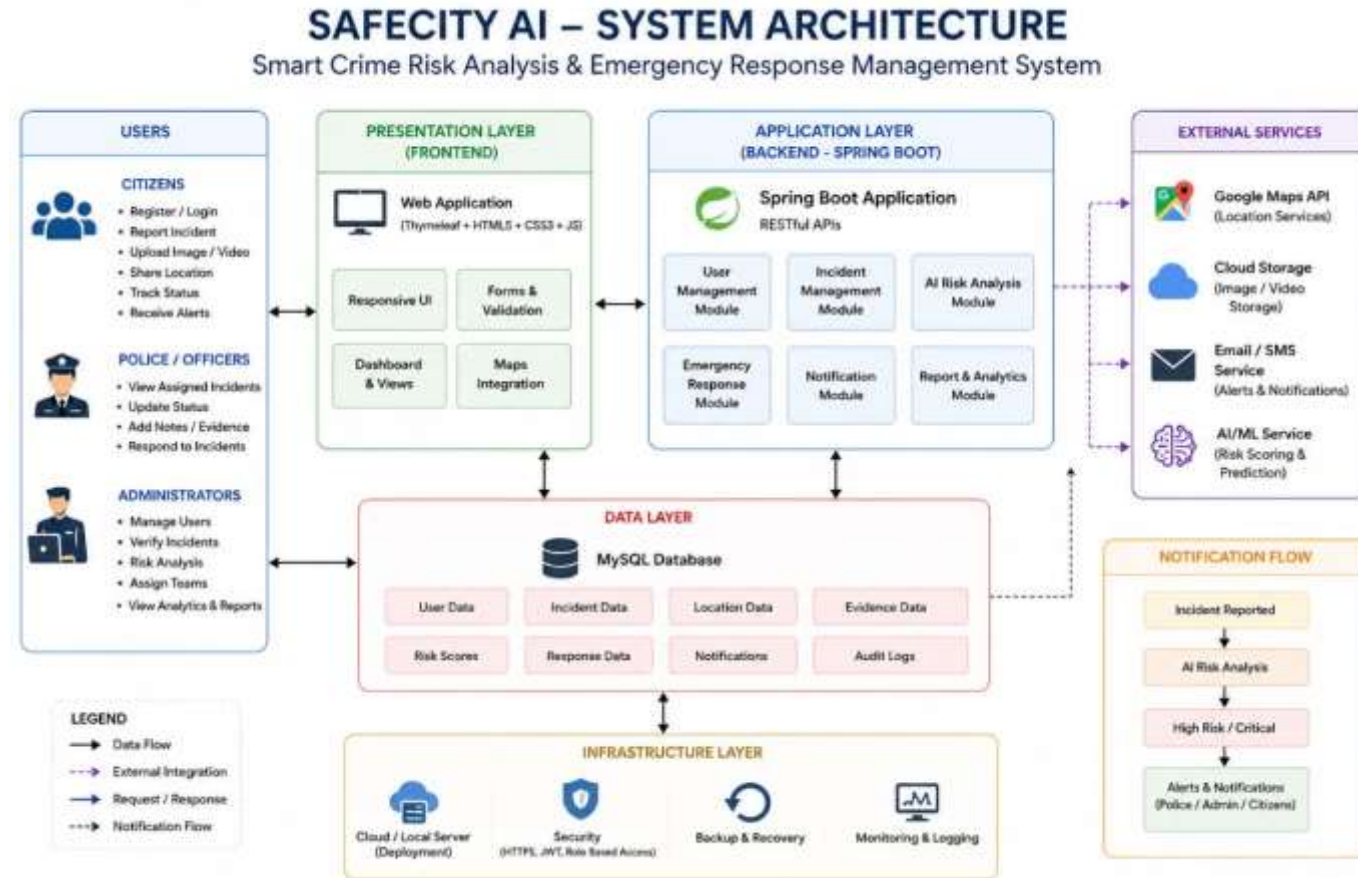
- Existing security systems mainly depend on manual CCTV monitoring.
- Crime-related data is not effectively used to predict high-risk areas.
- Suspicious activities may not be detected in real time.
- Emergency alerts and police response can be delayed.
- Therefore, an AI-based system is needed to predict crime, detect suspicious activities, and provide faster emergency response.



# Objectives

- Predict possible crime locations using historical and real-time data.
- Detect suspicious activities through AI-based video analysis.
- Alert police and emergency services when a threat is detected.
- Help authorities respond quickly and manage emergencies effectively.

# System Architecture





# Module 1: Crime Data Management

- Collects and validates crime reports submitted by citizens through the website using Java input validation and conditional statements.
- Stores verified crime records in the MySQL database using Java JDBC and CRUD operations.
- Allows administrators to review, approve, update, or reject reports using OOP, methods, and event handling.
- Maintains a reliable historical crime database for further analysis using MySQL, collections, and exception handling.



# Report a Crime

Submitted reports are stored as Pending until an admin verifies them.

LOCATION

e.g. Anna Nagar

CRIME TYPE

Theft



Submit Report

## Recent Submissions

| ID   | LOCATION   | CRIME     | TIME                  | STATUS   |
|------|------------|-----------|-----------------------|----------|
| #107 | T Nagar    | Robbery   | 10 Aug 2026, 05:50 pm | Pending  |
| #108 | Velachery  | Assault   | 10 Aug 2026, 03:50 pm | Verified |
| #109 | Tambaram   | Vandalism | 09 Aug 2026, 09:50 pm | Verified |
| #106 | Anna Nagar | Theft     | 09 Aug 2026, 09:30 pm | Verified |
| #162 | Velachery  | Assault   | 08 Aug 2026, 09:50 pm | Verified |
| #166 | Tambaram   | Vandalism | 08 Aug 2026, 09:50 pm | Verified |
| #163 | Velachery  | Assault   | 07 Aug 2026, 09:50 pm | Verified |



# Module 2: AI-based Crime Pattern Analysis And Risk Prediction

- Analyzes crime records based on location, crime type, date, and time using Java loops, collections, and date/time handling.
- Calculates a risk score for an area using Java-based rule-based conditions and arithmetic operations.
- Classifies areas as Low, Medium, or High Risk using conditional statements and decision-making logic.
- Identifies crime-prone areas from historical records using JDBC, OOP, methods, and collections.



## MODULE 02

# AI Crime Risk Prediction

Analyses historical crime data to estimate risk for an area.

LOCATION

Anna Nagar



CRIME TYPE

Theft

**Predict Risk**

## How the prediction works

The predictor counts how many past reports match the chosen location and crime type, then classifies the risk:

- **High** — 15+ matching cases, or an area with heavy overall crime
- **Medium** — 5-14 matching cases
- **Low** — fewer than 5 matching cases

EVERY RESULT INCLUDES A REASON — NOT JUST A SCORE

PREDICTIONS COMPUTED LIVE FROM THE SEEDDED CRIME DATASET

# Module 3: Intelligent Emergency Response And Decision Support

- Generates emergency alerts based on the identified risk level using Java conditional statements and decision-making logic.
- Displays police stations and emergency contact information using JDBC, MySQL, and collections.
- Provides safety recommendations according to the crime risk level using methods and rule-based logic.
- Updates the emergency response dashboard with current crime and risk information using OOP, JDBC, and exception handling.



# Emergency Response

Automatic alerts, safety recommendations, and police contacts for high-risk areas.

## Active High-Risk Alerts

### ⚠ High Risk — Anna Nagar

Total reports: 39  
Nearest station: Anna Nagar Police Station  
Emergency number: 100 / 112  
Recommendation: Increase police patrol and remain alert.

## Police Stations & Emergency Contacts

| AREA       | POLICE STATION            | CONTACT   |
|------------|---------------------------|-----------|
| Anna Nagar | Anna Nagar Police Station | 100 / 112 |
| T Nagar    | T Nagar Police Station    | 100 / 112 |
| Velachery  | Velachery Police Station  | 100 / 112 |
| Tambaram   | Tambaram Police Station   | 100 / 112 |
| Adyar      | Adyar Police Station      | 100 / 112 |

National emergency numbers: 100 / 112

## Response note

This system only displays emergency information and recommendations for the project demo. In a real smart-city deployment, this module could integrate with police and emergency services to send alerts automatically.

# Advantages

- Detects suspicious activities early and helps prevent potential crimes.
- Sends instant alerts to police and emergency services for faster response.
- Uses AI to identify high-risk areas and support preventive action.
- Provides 24/7 smart monitoring to continuously detect potential threats.



# Conclusion

- Safe City AI uses Artificial Intelligence to analyze crime data and predict possible crime-prone areas.
- The system uses real-time CCTV and sensor data to identify suspicious activities and potential threats.
- When a threat is detected, it automatically generates alerts and notifies the police or emergency services.
- By providing faster detection, prediction, and emergency response, the system helps improve public safety and create a safer smart city.





*Thank You*